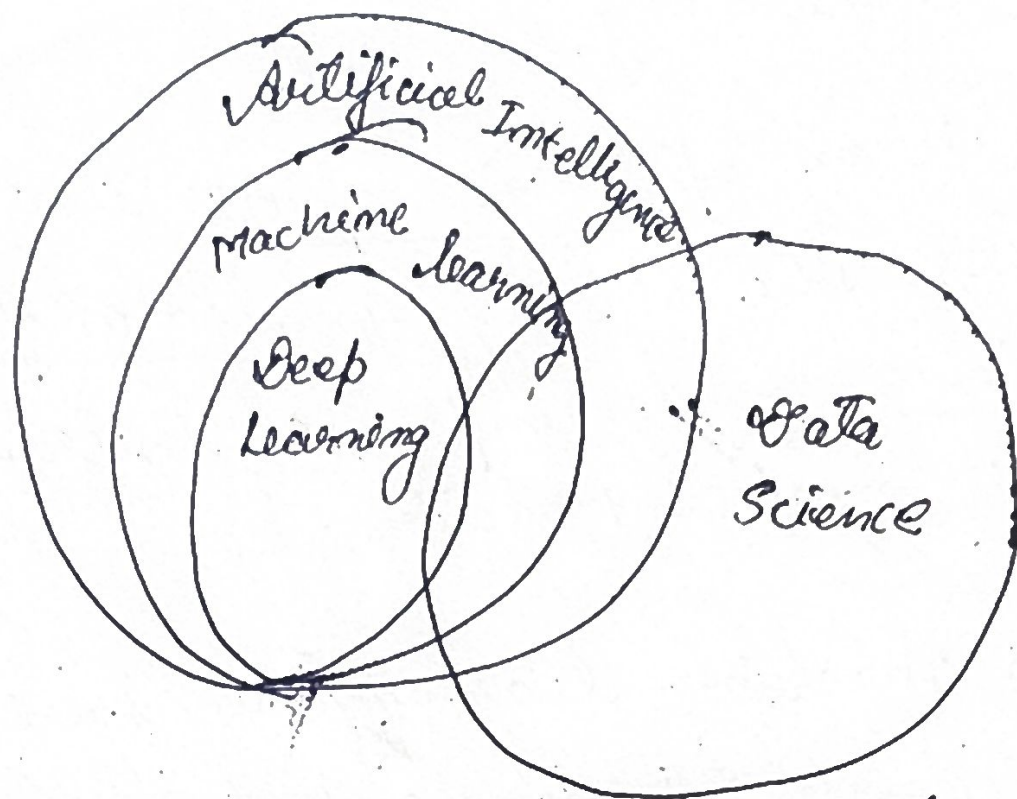


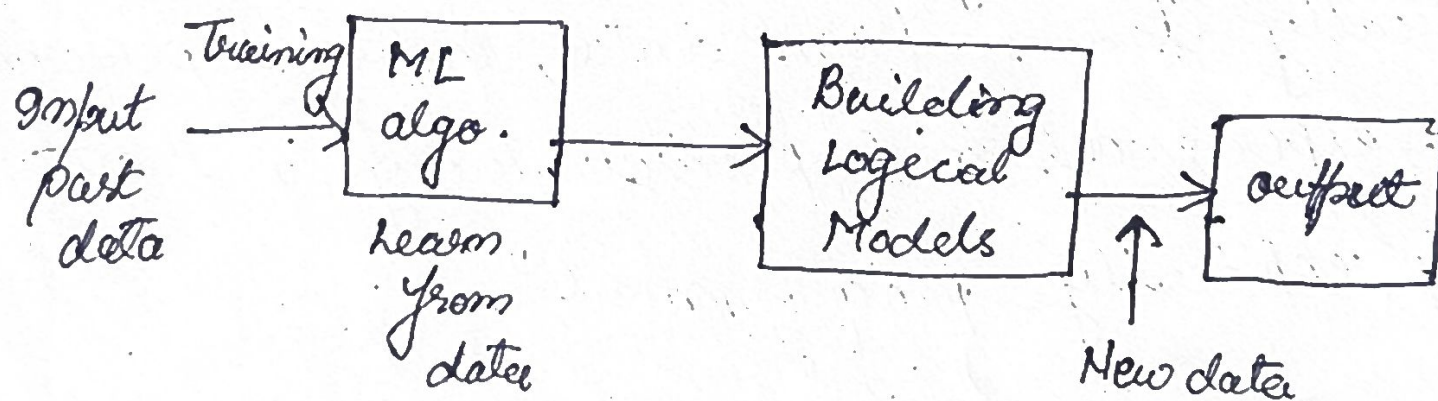
Machine Learning:

- ML is an application of AI that gives devices the ability to learn from their experience and improve their self without any human interaction.
- Thus machine learning is said as subject of artificial intelligence that is mainly concerned with the development of algorithm which allow a computer to learn the data and past experiences on their own. The term machine learning was first introduced by "Arthur Samuel in 1959". we can it in summarized way as:
- "Machine learning enables a machine to automatically learn from data, improve performance from experiences and predict things without being explicitly programmed".



How does machine learning work.

- A machine learning system learns from historical data, builds the prediction models and whenever it receives new data, predicts the output.
- ~~The~~ The accuracy of predicted output depends upon the amount of data, as the huge amount of data helps to build a better model which predicts the output more accurately.



Feature of machine learning.

- ↳ Machine learning uses data to detect various pattern in given dataset.
- ↳ It can learn from past data and improve automatically.
- ↳ It is data-driven technology.
- ↳ Machine learning is much similar to data mining as it also deals with the huge amount of the data.

Need / Importance of machine learning.

- ↳ ML is important because it gives enterprises a view of trends in customer behavior and business operational pattern, as well as supports the development of new products.
- ↳ Need The reason behind the need for machine learning is that it is capable of doing tasks that are too complex for a person to implement directly.
- ↳ We can train machine learning algorithms by providing them the huge amount of data and let them explore the data construct the models and predict the required output automatically.
- ↳ Thus solving complex problems, which are difficult.

for a human.

↳ Decision making in various sector like
finance, self-driving car, cyber fraud detection
face recognition

↳ finding hidden patterns and extracting useful
information from data.

Type of machine Learning 3

→ These ML algorithms help to solve different business problems like regression, classification, forecasting, clustering and association etc.

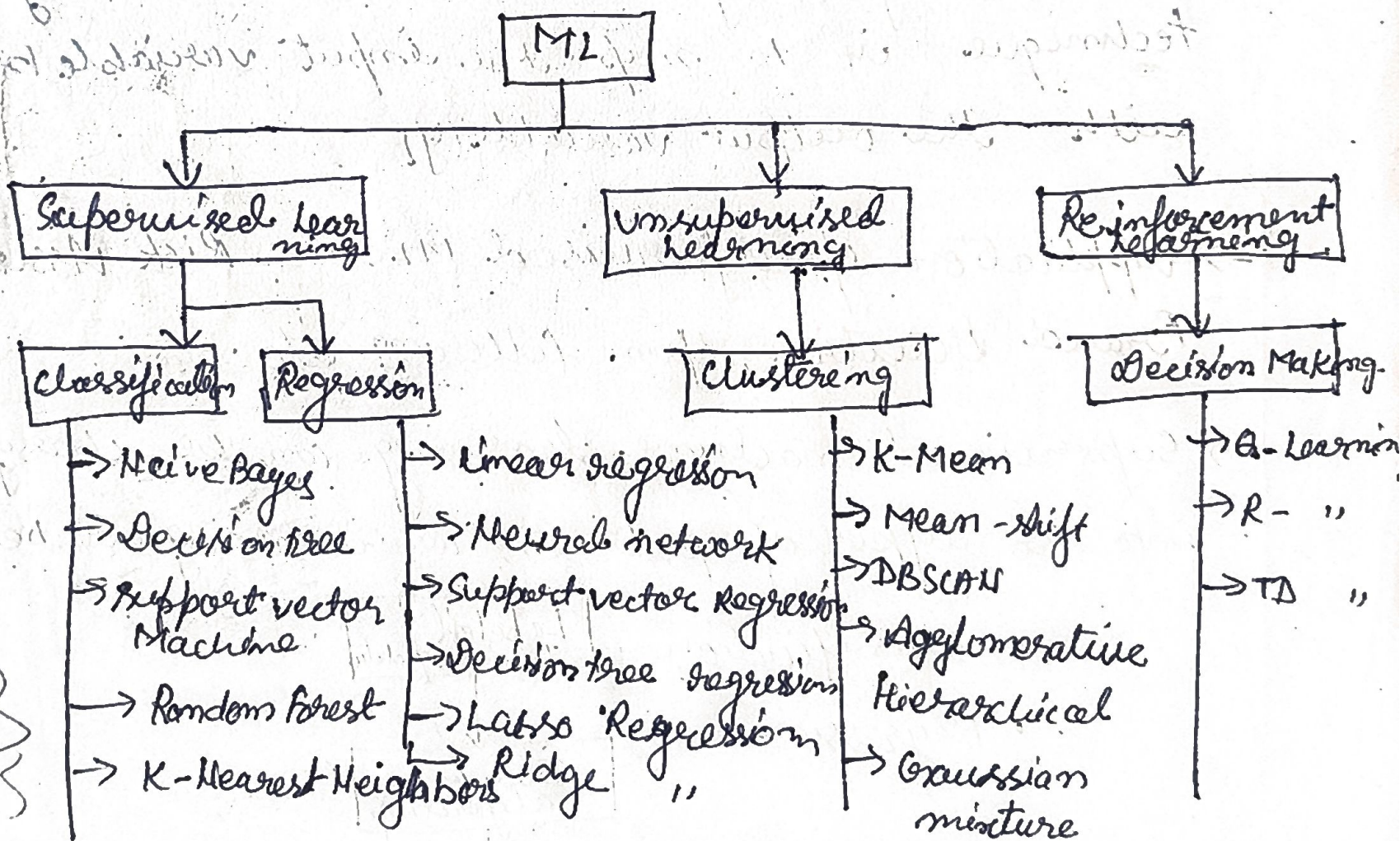
→ Based on the method and way of learning Machine learning is divided into mainly four types

- Supervised machine learning

- Unsupervised

- Semi-Supervised

- Reinforcement



Supervised Machine Learning

→ It is based on supervision.

→ we train the machine using the labelled database

→ Based on training, the machine predicts the output.

→ Here labelled data specifies that some of the inputs are already mapped to the output.

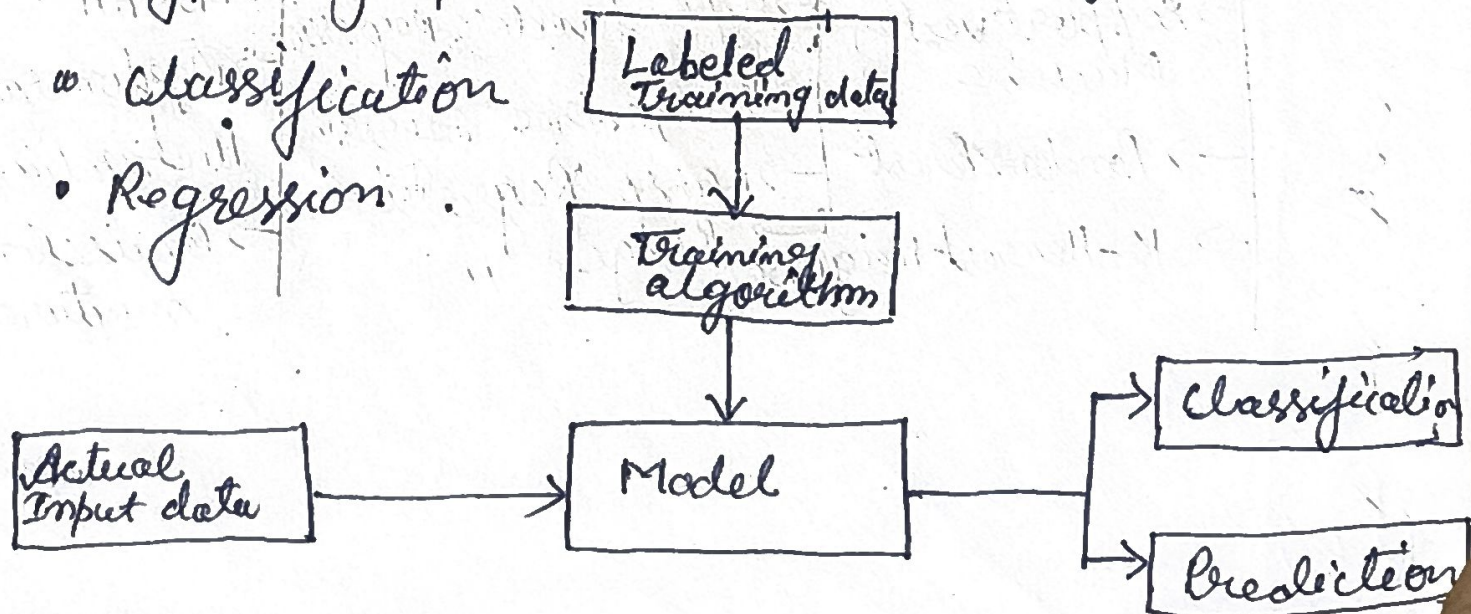
→ First we train the machine with the ip and corresponding output and then we ask the machine to predict the output using test dataset.

→ The main goal of the supervised learning technique is to map the input variable (x) with the output variable (y)

→ Application of supervised ML are Risk Assessment, Fraud Detection, spam filtering etc.

→ Supervised machine learning can be classified into two types of problems, which are given below

- Classification
- Regression



Classification

→ Classification algorithms are used to solve the classification problems in which the output variable is categorical such as "yes" or "no", "Male" or "Female", etc.

Binary / multi class classification

→ Some popular classification algo are

- Random Forest
- Decision tree
- Logistic regression
- Support vector Machine algorithm

Regression:

→ These algorithms are used to solve regression problems in which there is a linear relationship b/w input and output variable.

→ These are used to predict continuous output variables, such as market trends, weather prediction etc.

→ Some popular regression algorithms are given below:

- Simple linear Regression algorithm.
- Multivariate " "
- Decision tree " "
- Lasso Regression.

Advantage of supervised learning

- Since supervised learning works with the labelled dataset so we can have an exact idea about the classes of object.
- These algorithms are helpful in predicting the output on the basis of prior experience.

Disadvantage.

- These algorithms are not able to solve complex task.
- It may predict the wrong output if the test data is different from the training data.
- It requires lots of computational time to train the algorithm.

Unsupervised machine learning

5

↳ Models are not supervised using training dataset.

↳ Model itself find the hidden patterns and insights from the given data.

↳ It can be compared to learning which takes place in human brain while learning new things.

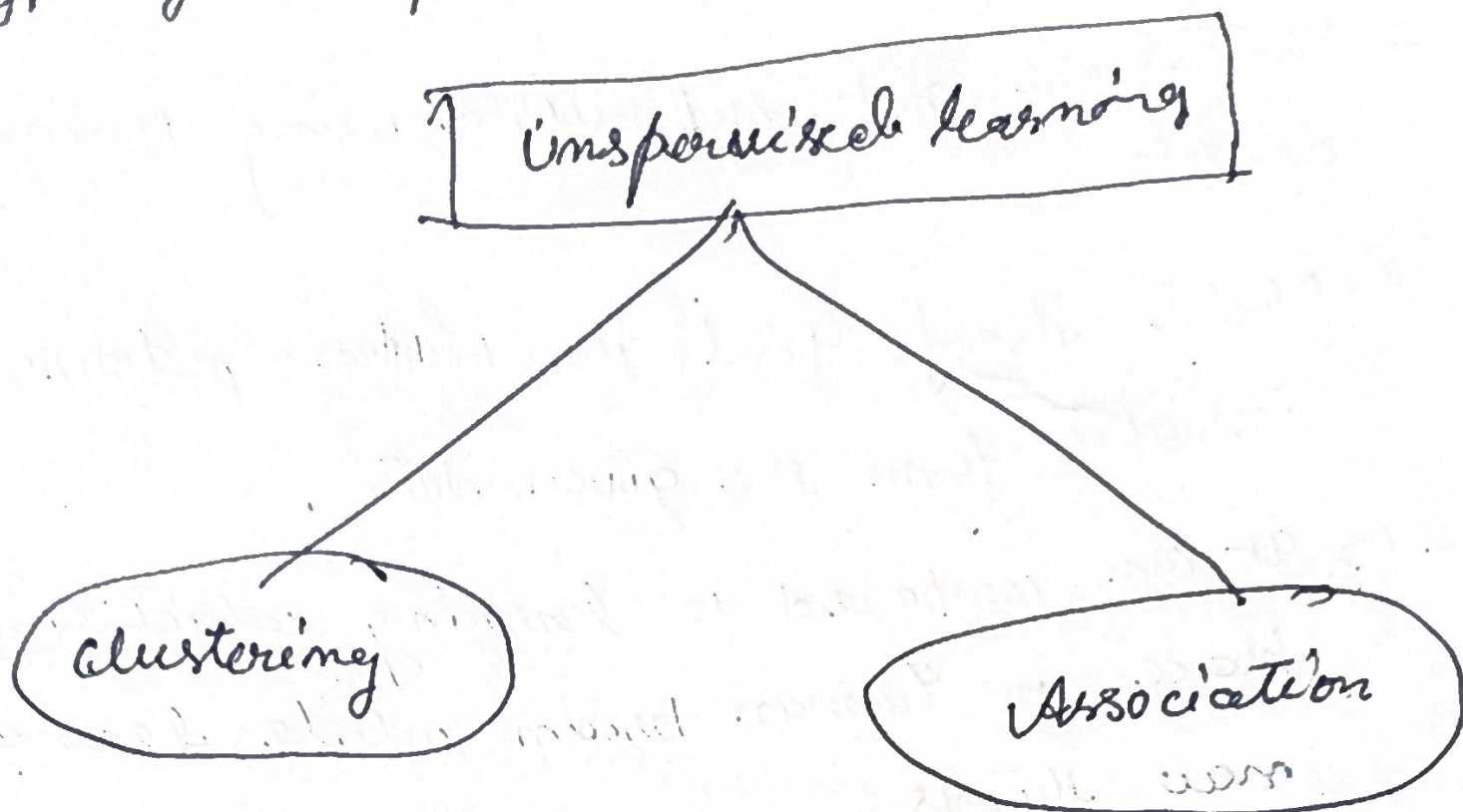
↳ It can not be applied to regression or classification problems.

↳ We have input data but no corresponding output data.

↳ Find the underlying structure of dataset, group that data according to similarities and represent

Definition: Unsupervised learning is a type of machine learning in which model are trained using unlabeled dataset and are allowed to act on that data without any supervision.

Type of Unsupervised learning.



clustering:

↳ It is a method of grouping the objects into clusters such that object with most similarities remains into group and has less or no similarities with the objects of another group.

↳ It finds the commonalities between the data objects and categorizes them as per the ~~per~~ presence and absence of those commonalities.

Association:

↳ It is ~~not~~ rule that is an unsupervised learning method which is used for finding the relationships between variables in the large database.

↳ It determines the set of items that occurs together in the dataset.

↳ Association rule makes marketing strategy more effective

Ex Bread $\rightarrow$ butter/jam

Unsupervised learning algorithms:

↳ k-mean clustering

↳ K-MN

↳ Hierarchical clustering

↳ Anomaly detection

↳ Neural networks

↳ Principal component Analysis

↳ Apriori algo.

Advantage:

↳ It is used for more complex tasks as compared to supervised learning

↳ It is preferable as it is easy to get unlabeled data in comparison to labeled data

Disadvantage:

↳ It is intrinsically more difficult than supervised learning as it does ~~not~~ not have corresponding output.

↳ It might be less accurate as if data is not labeled and algo do not know the exact output in advance.

Reinforcement learning:

↳ It is machine learning training method based on rewarding desired behaviors and/or punishing undesired ones.

↳ A reinforcement learning agent is able to perceive and interpret its environment, take actions and learn through trial and errors.

↳ It is feedback-based machine learning technique in which an agent learn to behave in an environment by performing the action and seeing the result of action. For each good action, the agent gets positive feedback and for each bad action, the agent get negative feedback or penalty.

↳ Since there is no labeled data, so the agent is bound to learn by its experience only.

↳ RL solves a specific type of problems where decision making is sequential and goal of an agent is in long terms such as game-playing, robotics.

⑫
⇒ Reinforcement learning is sub-branch of machine learning that trains a model to return an optimal solution for a problem by taking a sequence of decisions by itself.

→ we model an environment after the problem statement. The model interacts with this environment and comes up with solution all on its own, without human interference.

→ To push it in right direction, we simply give it a positive reward if it performs an action that brings it closer to its goal or a negative reward if it goes away from its goal.

⇒ Need for Deep Reinforcement Learning.

↳ tremendous amount of data is needed to train model.

↳ The more complex a model, the more data it may require. But this data may not be available to us. Data collected might not be reliable.

↳ Learning from a small subset of action will not help expand the vast realm of solution that may work for a particular problem.

↳ All of these problems are overcome by deep reinforcement learning.

↳ Thus, in deep reinforcement learning, we introduce our model to controlled environment which is modeled after the problem statement.

Reinforcement learning ^{Type:} ~~working~~ 8

↳ 1. Positive:

Positive reinforcement is defined as when an event occurs due to a particular behaviour, increases the strength and frequency of the behaviour. In other words, it has a positive effect on behaviour.

2. Negative:

Negative reinforcement is defined as strengthening of behavior because a negative condition is stopped or avoided.

Terms used in Reinforcement learning:

- **Agent():** An entity that can perceive/explore the environment and act upon it.
- **Environment():** A situation in which an agent is present, which means it is a situation in nature.
- **Action():** Action are the moves taken by an agent within the environment.
- **State():** State is a situation returned by environment after each action taken by the agent.
- **Reward():** A feedback returned to the agent from the environment to evaluate the action of the agent.
- **Policy():** Policy ^(π) is strategy applied by the agent for next action based on the current state.

value(): It is expected long-term returned with discount factor and opposite to the short-term reward.

Q-value(): It is mostly similar to the value, but it takes one additional parameter as a current action

Approaches to implement Reinforcement Learning

There are mainly three ways to implement reinforcement learning in RL, which are;

1. Value Based:

The value-based approach is about to find the optimal value function, which is the maximum value at a state under any policy. Therefore, the agent expects the long-term return at any state under policy.

2. Policy - Based

Policy-based approach is to find the optimal policy for maximizing future rewards without using the value function.

In this approach, the agent tries to apply such a policy that the action performed in each step helps to maximize the future reward.

The policy-based approach has mainly two types of policy at any state

- Deterministic: The same action is produced by the policy at any state.

- Stochastic: In this policy, probability determines the produced action.

Model-based :

In the model based approach, a virtual model is created for environment, and the agent explores the environment to learn it. There is no particular solution or algorithm for this approach because the model representation is different for each environment.

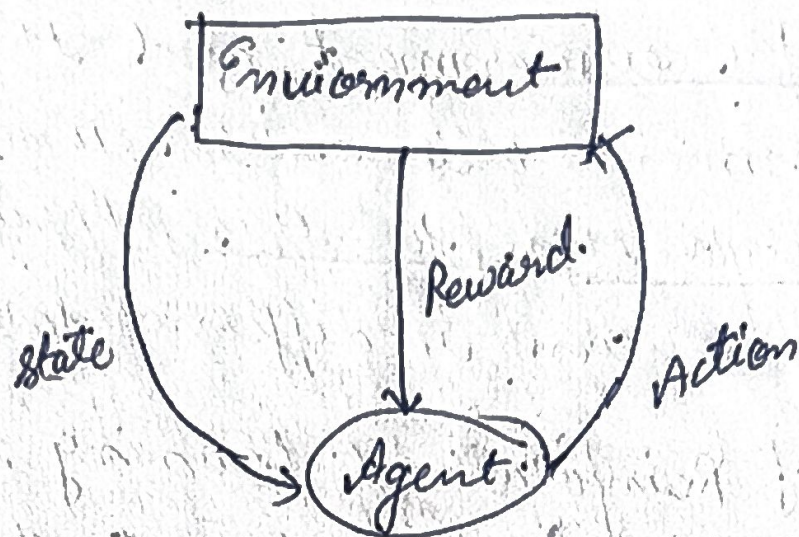
Element of Reinforcement Learning.

There are four main elements of Reinforcement Learning :

1. Policy
2. Reward Signal
3. Value function
4. Model of the environment.

Application of Reinforcement Learning :

1. Robotics: Robotics with pre-programmed behavior are useful in structured environments such as the assembly line of an automobile manufacturing plants, where the task is repetitive in nature.
 - ↳ A master chess player makes a move. The choice is informed both by playing, anticipating possible replies and counter replies.
 - ↳ Problem like object detection, face recognition, fraud detection can be better classifier than by RF



1. Policy

2. Reward

3. Value function

4. Model

Application of Reinforcement Learning

1. Robotics: Reinforcement learning is used to train robots to perform tasks by rewarding them for successful actions and punishing them for failures.

2. Game playing: Reinforcement learning is used to train agents to play games like Go, Chess, and StarCraft.

3. Recommendation systems: Reinforcement learning is used to recommend products or services to users based on their past behavior.

4. Natural language processing: Reinforcement learning is used to train models to generate human-like text.

5. Healthcare: Reinforcement learning is used to optimize treatment plans for patients.